



NSW Education Standards Authority

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Centre Number

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Student Number

2024 HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Write your Centre Number and Student Number at the top of this page

Total marks: 100

Section I – 10 marks (pages 2–8)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 9–39)

- Attempt Questions 11–31
- Allow about 2 hours and 45 minutes for this section

Section I

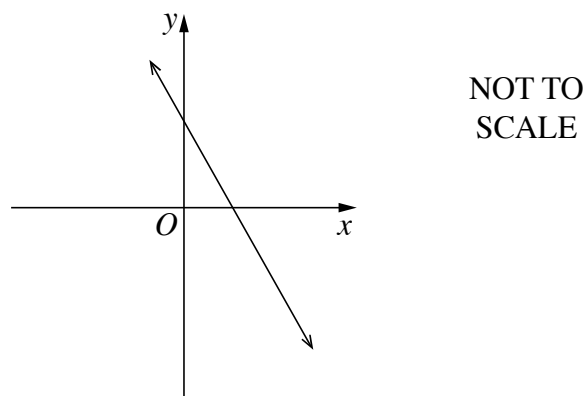
10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 Consider the function shown.



Which of the following could be the equation of this function?

- A. $y = 2x + 3$
 - B. $y = 2x - 3$
 - C. $y = -2x + 3$
 - D. $y = -2x - 3$
- 2 In a group of 60 students, 38 play basketball, 35 play hockey and 5 do not play either basketball or hockey.

How many students play both basketball and hockey?

- A. 55
- B. 18
- C. 13
- D. 8

- 3 Pia's marks in Year 10 assessments are shown. The scores for each subject were normally distributed.

	<i>Pia's mark</i>	<i>Year 10 mean</i>	<i>Year 10 standard deviation</i>
English	78	66	6
Mathematics	80	71	10
Science	77	70	15
History	85	72	9

In which subject did Pia perform best in comparison with the rest of Year 10?

- A. English
 - B. Mathematics
 - C. Science
 - D. History
- 4 The parabola $y = (x - 3)^2 - 2$ is reflected about the y -axis. This is then reflected about the x -axis.

What is the equation of the resulting parabola?

- A. $y = (x + 3)^2 + 2$
- B. $y = (x - 3)^2 + 2$
- C. $y = -(x + 3)^2 + 2$
- D. $y = -(x - 3)^2 + 2$

5 What is $\int (6x + 1)^3 dx$?

A. $\frac{1}{24}(6x + 1)^4 + C$

B. $\frac{1}{4}(6x + 1)^4 + C$

C. $\frac{2}{3}(6x + 1)^4 + C$

D. $\frac{3}{2}(6x + 1)^4 + C$

6 What is the domain of the function $f(x) = \frac{1}{\sqrt{x^2 - 1}}$?

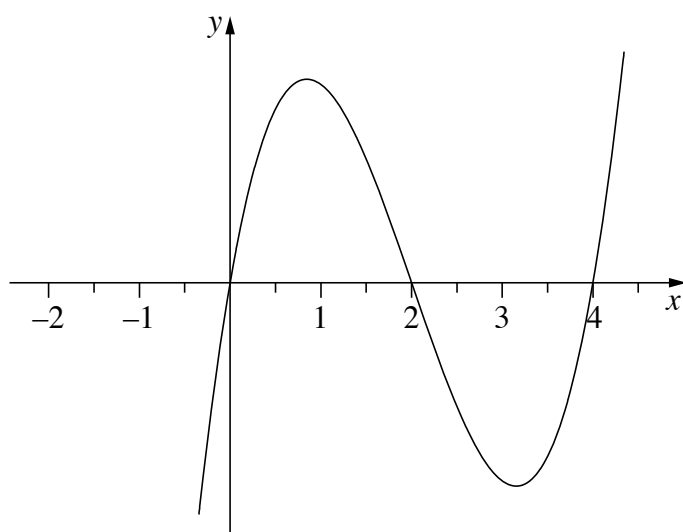
A. $[-1, 1]$

B. $(-\infty, -1] \cup [1, \infty)$

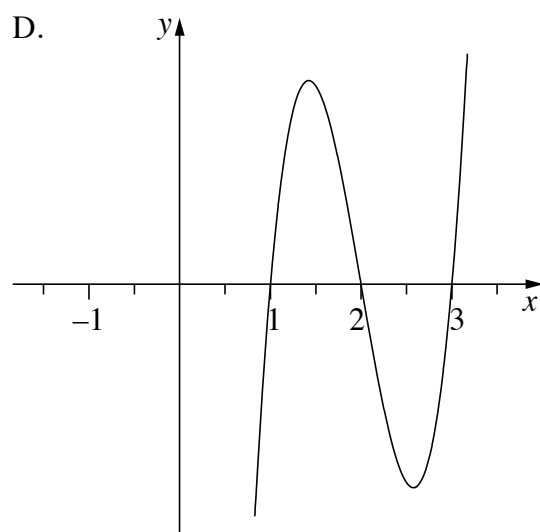
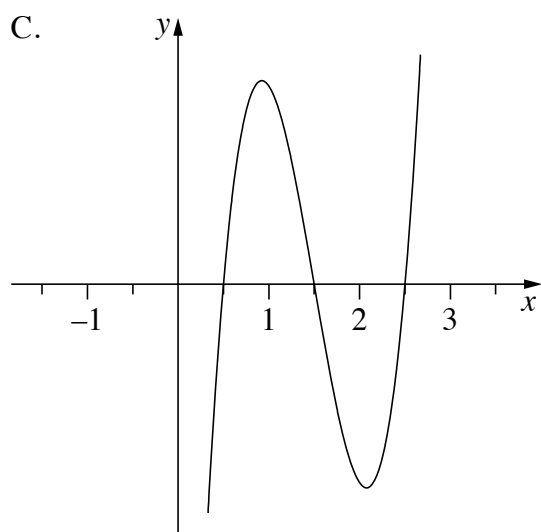
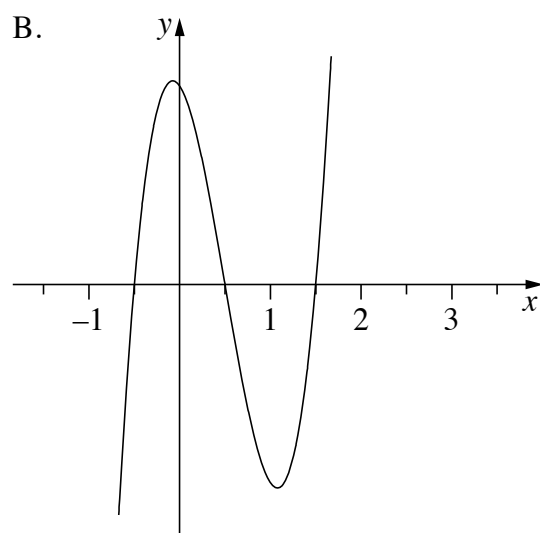
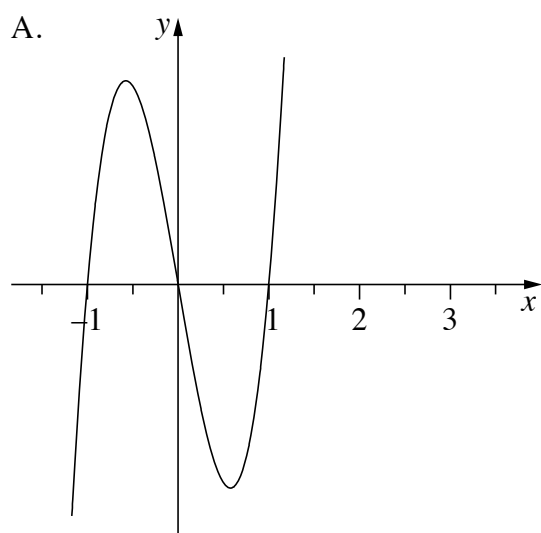
C. $(-1, 1)$

D. $(-\infty, -1) \cup (1, \infty)$

- 7 The diagram shows the graph $y = f(x)$.



Which of the following best represents the graph $y = f(2x - 1)$?

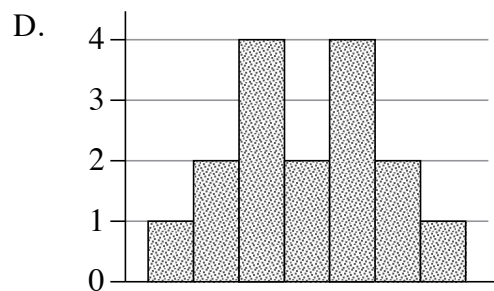
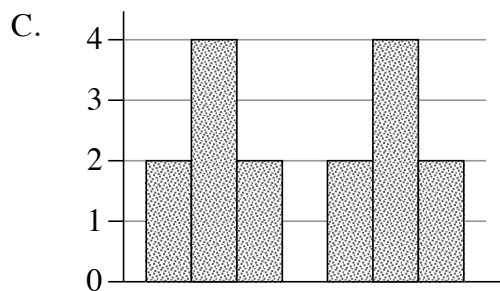
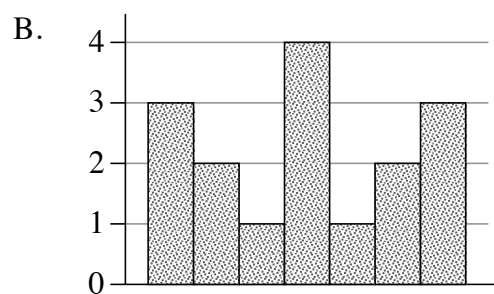
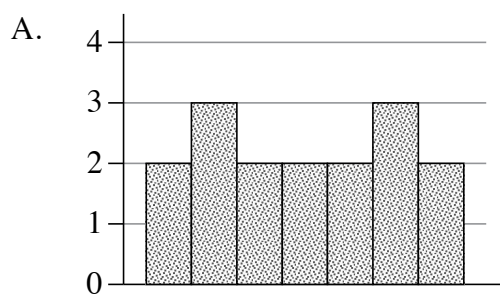


- 8 Some data are used to create a box plot shown.

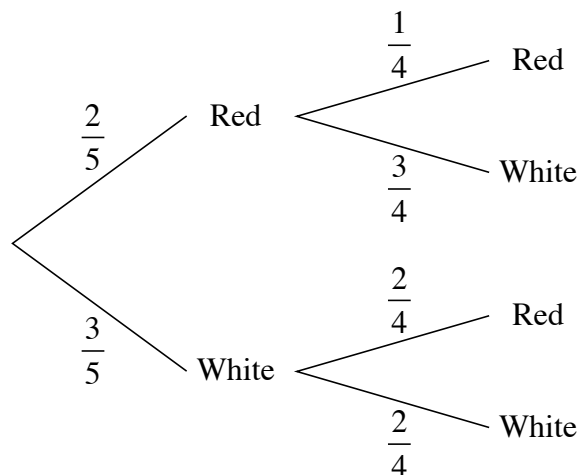


A histogram is created from the same set of data.

Which of these histograms is NOT possible for the given box plot?



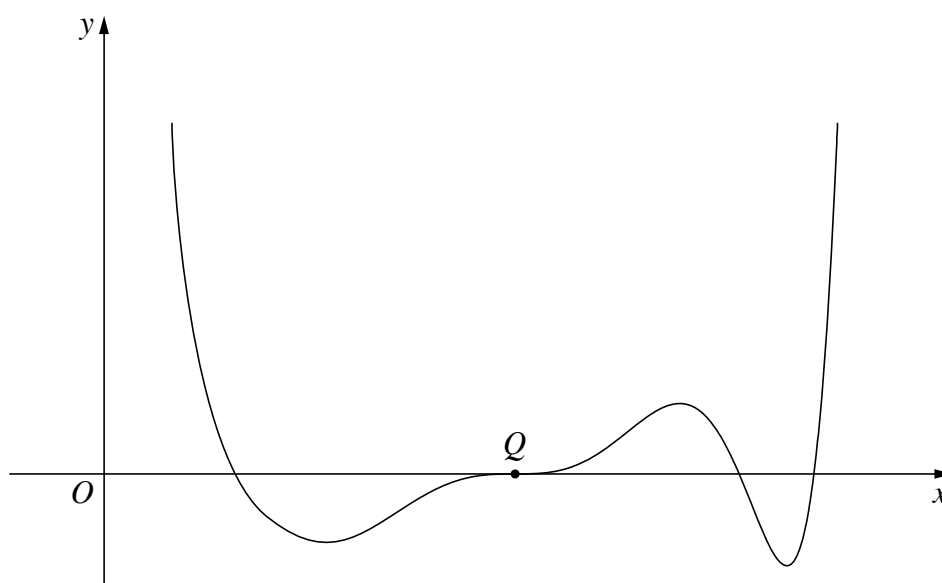
- 9 A bag contains 2 red and 3 white marbles. Jovan randomly selects two marbles at the same time from this bag. The probability tree diagram shows the probabilities for each of the outcomes.



Given that one of the marbles that Jovan has selected is red, what is the probability that the other marble that he has selected is also red?

- A. $\frac{1}{10}$
B. $\frac{1}{7}$
C. $\frac{1}{4}$
D. $\frac{7}{10}$

- 10 The diagram shows the graph $y = f(x)$.



The point Q is a horizontal point of inflection.

Let $A(x) = \int_0^x f(t) dt$.

How many points of inflection does the graph $y = A(x)$ have?

- A. 2
- B. 3
- C. 4
- D. 5